Functions & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

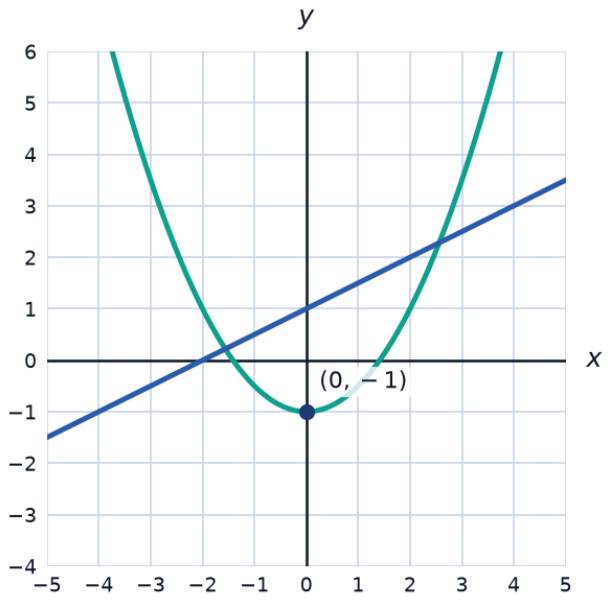
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Function transformations

Let $f(x) = x^2$ and $g(x) = -3f(x+2) - 4 = -3(x+2)^2 - 4$. Find $g(0)$, and describe the transformation from f to g in words.

2 [Foundational] • Target: Function transformations

Using the same $g(x) = -3(x+2)^2 - 4$, find $g(-2)$, and explain why this is the vertex of g's graph.



3 [Foundational] • Target: Even and odd functions

Classify $f(x) = x^4 + 2x^2 - 5$ as even, odd, or neither.

4 [Foundational] • Target: Even and odd functions

Classify $h(x) = 2x^5 - 3x$ as even, odd, or neither.

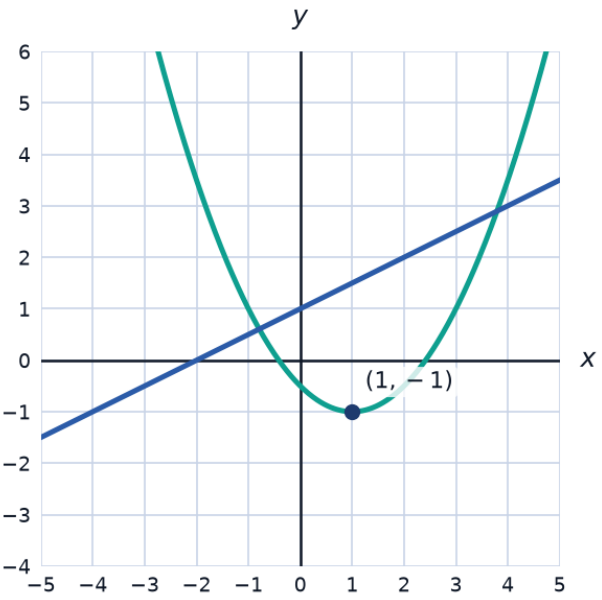
5 [Foundational] • Target: Even and odd functions

Explain why $k(x) = x^2 + x$ is neither even nor odd.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Domain restrictions

Find the domain of $f(x) = \sqrt{2x + 6}$.



7 [Foundational] • Target: Domain restrictions

Find the domain of $g(x) = \frac{1}{x^2 - 25}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

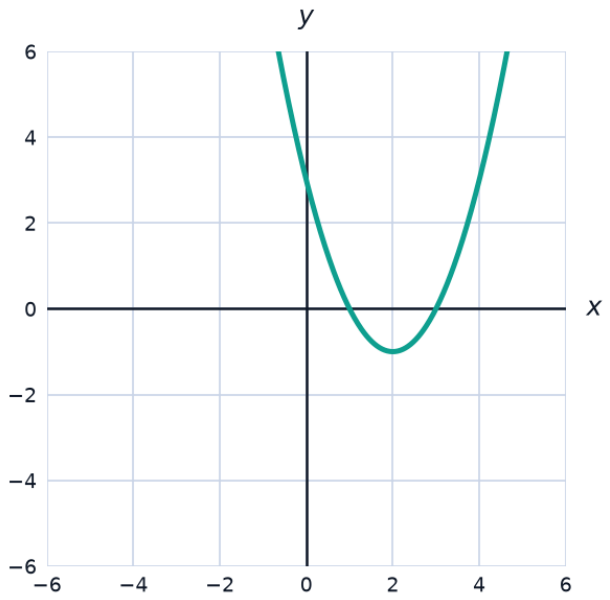
8 [Intermediate] • Target: Piecewise functions

Let $p(x) = 2x + 3$ for $x < 1$, and $p(x) = x^2 - 1$ for $x \geq 1$. Find $p(-2)$ and $p(3)$.

© ClipSAT — clipsat.org | All Rights Reserved

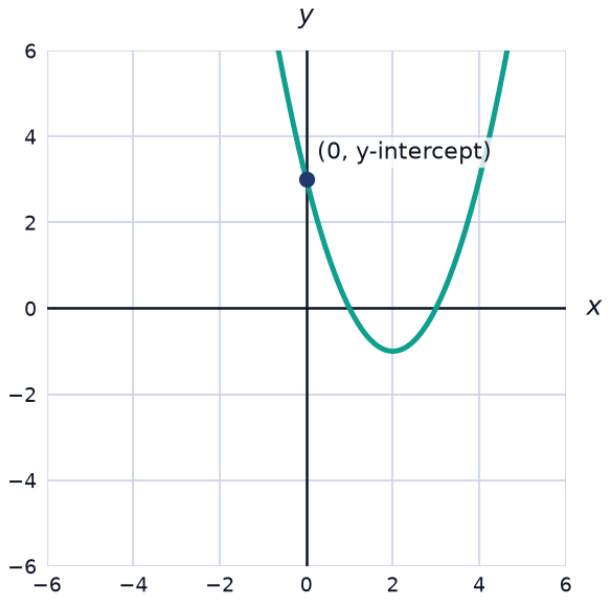
9 **[Intermediate]** • **Target: Function transformations**

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



10 **[Intermediate]** • **Target: Even and odd functions**

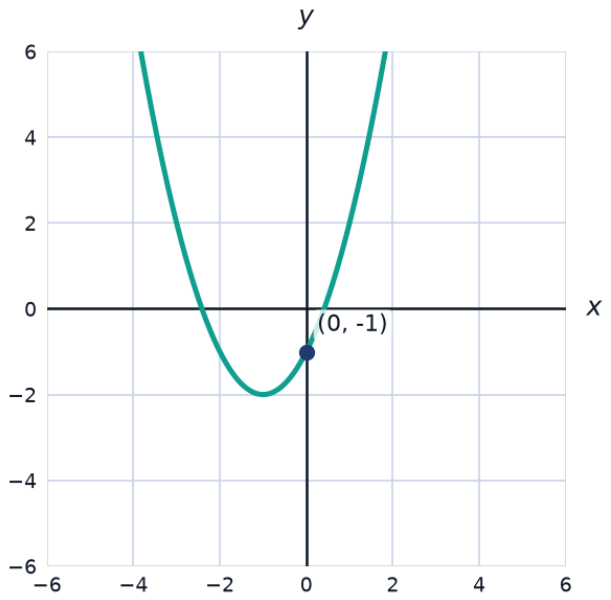
Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?



© ClipSAT — clipsat.org | All Rights Reserved

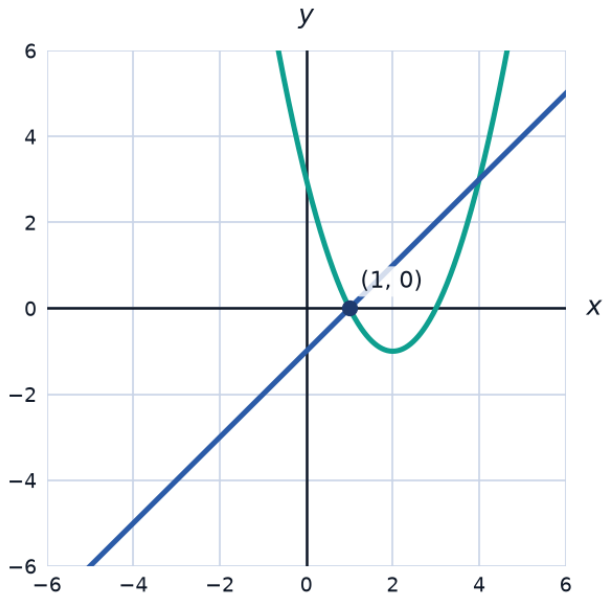
11 [Intermediate] • Target: Domain restrictions

Consider the graph of the function $f(x)$ shown below. What is the value of $f(0)$?



12 [Intermediate] • Target: Piecewise functions

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?



13 [Intermediate] • Target: Rational Functions & Oblique Asymptotes

Consider the function $f(x) = x^3 - 6x^2 + 5$. A tangent line to the curve of $f(x)$ is drawn at the point $P(a, f(a))$. This tangent line also intersects the curve at another point $Q(b, f(b))$. If the x-coordinate of the point P is $a = 1$, determine the x-coordinate of the point Q.

14 [Intermediate] • Target: Piecewise & Absolute Value Graphs

Find the domain of $f(x) = \sqrt{x - 9} + \frac{1}{x - 11}$.

© ClipSAT — clipsat.org | All Rights Reserved

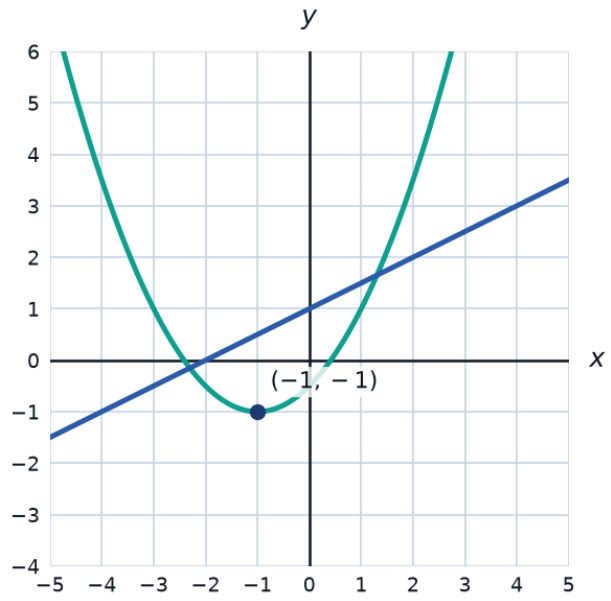
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Inverse Trigonometric Graphs & Domains** • **Cross-Lesson: 2.01 Sequences & Series**

A manufacturer produces light bulbs. The probability that a randomly selected light bulb is defective is p. The production process is designed such that the probability of producing exactly k defective bulbs in a batch of $n = 5$ bulbs follows the binomial probability formula: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. It is known that the probability of finding exactly 2 defective bulbs in a batch of 5 is $\frac{10}{32}$. Find the value of p.

16 **[Challenging]** • **Target: Transformations of Non-Linear Functions** • **Cross-Lesson: 4.01 Matrices & Vectors**

Consider the transformation grid below. If point A has coordinates $(1, 2)$, what are the coordinates of its image, A', after the transformation shown?



17 **[Challenging]** • **Target: Function transformations** • **Cross-Lesson: 5.01 Introductory Calculus**

A factory produces two types of components, A and B. Components of type A have a defect rate of 3%, and components of type B have a defect rate of 5%. A batch of 100 components is produced, consisting of N components of type A and $100 - N$ components of type B. Let X be the random variable representing the total number of defective components in the batch. If the expected number of defective components is $E[X] = 4.2$, find the probability that a randomly selected component from this batch is defective, given that it is of type A.

18 **[Challenging]** • **Target: Even and odd functions** • **Cross-Lesson: 6.01 Probability**

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Domain restrictions • Cross-Lesson: 1.01 Complex Numbers

A matrix $M = (k, 2; 3, 1)$ transforms a point $P(1, 2)$ to $P'(5, 5)$. Find the value of k.

20 [Challenging] • Target: Piecewise functions • Cross-Lesson: 2.01 Sequences & Series

A bag contains 5 red marbles and 3 blue marbles. If a marble is drawn at random, what is the probability that it is blue?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Function transformations

Classify $f(x) = 2x^3 + 5$ as even, odd, or neither.

a. even

b. both even and odd

c. neither

d. odd

22 [Foundational] • Target: Even and odd functions

Classify $f(x) = -3x^5 + 6x$ as even, odd, or neither.

a. both even and odd

b. even

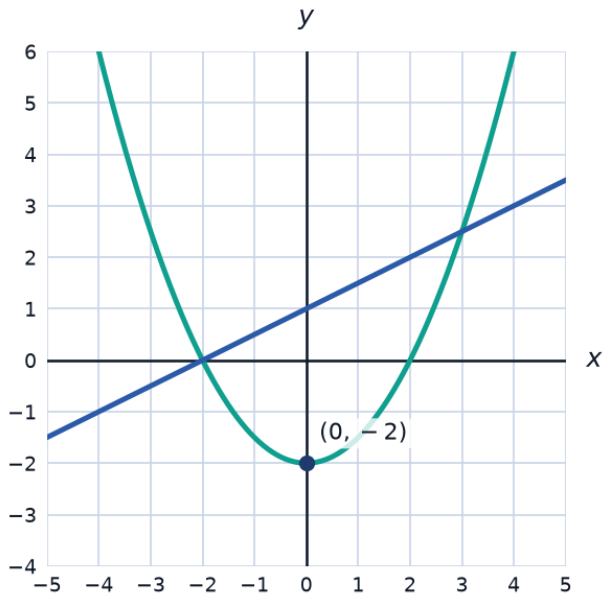
c. neither

d. odd

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Domain restrictions

Classify $f(x) = -x^2 + 6$ as even, odd, or neither.



a. both even and odd

b. neither

c. odd

d. even

24 [Foundational] • Target: Piecewise functions

Classify $f(x) = 3x^5 + 6x$ as even, odd, or neither.

a. even

b. neither

c. odd

d. both even and odd

25 [Foundational] • Target: Rational Functions & Oblique Asymptotes

Classify $f(x) = -2x^3 + 8$ as even, odd, or neither.

a. neither

b. odd

c. both even and odd

d. even

26 [Foundational] • Target: Piecewise & Absolute Value Graphs

Classify $f(x) = -3x^3 + 4x$ as even, odd, or neither.

a. even

b. neither

c. both even and odd

d. odd

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Inverse Trigonometric Graphs & Domains

Classify $f(x) = -x^4 + 8x^2$ as even, odd, or neither.

a. odd

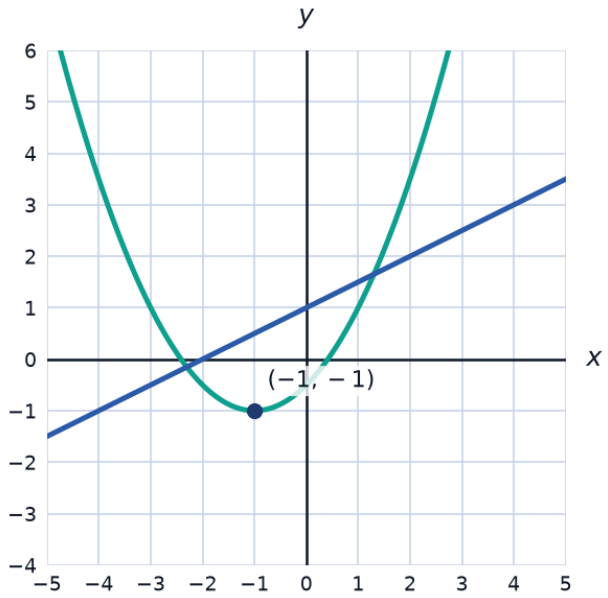
b. both even and odd

c. even

d. neither

28 [Foundational] • Target: Transformations of Non-Linear Functions

Find the domain of $f(x) = \sqrt{5x - 35}$.



a. $x \geq 9$

b. $x \geq 7$

c. $x \geq 5$

d. $x \leq 7$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Function transformations

Find the domain of $f(x) = \sqrt{4x - 24}$.

a. $x \geq 4$

b. $x \leq 6$

c. $x \geq 8$

d. $x \geq 6$

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Even and odd functions

Find the domain of $f(x) = \sqrt{x+3}$.

a. $x \leq -3$

b. $x \geq -5$

c. $x \geq -3$

d. $x \geq -1$

31 [Intermediate] • Target: Domain restrictions

Find the domain of $f(x) = \sqrt{5x+25}$.

a. $x \geq -5$

b. $x \geq -7$

c. $x \leq -5$

d. $x \geq -3$

32 [Intermediate] • Target: Piecewise functions

Find the domain of $f(x) = \sqrt{5x}$.

a. $x \geq 0$

b. $x \geq 2$

c. $x \leq 0$

d. $x \geq -2$

33 [Intermediate] • Target: Rational Functions & Oblique Asymptotes

Find the domain of $f(x) = \sqrt{-2x+12}$.

a. $x \leq 6$

b. $x \leq 8$

c. $x \leq 4$

d. $x \geq 6$

34 [Intermediate] • Target: Piecewise & Absolute Value Graphs

Find the domain of $f(x) = \sqrt{-3x-27}$.

a. $x \leq -11$

b. $x \geq -9$

c. $x \leq -7$

d. $x \leq -9$

35 [Intermediate] • Target: Inverse Trigonometric Graphs & Domains

Let $p(x) = 3x - 6$ for $x < 3$, and $p(x) = -2x + 3$ for $x \geq 3$. Find $p(-2)$ and $p(7)$.

a. $-12, -11$

b. $-12, -13$

c. $-10, -11$

d. $-11, -12$

© ClipSAT — clipsat.org | All Rights Reserved

36 [Intermediate] • Target: Transformations of Non-Linear Functions

Let $p(x) = 4x - 7$ for $x < 5$, and $p(x) = -x + 1$ for $x \geq 5$. Find $p(3)$ and $p(5)$.

a. $5, -6$

b. $5, -4$

c. $-4, 5$

d. $7, -4$

37 [Intermediate] • Target: Function transformations

Let $f(x) = x^2$ and $g(x) = 4f(x-3) - 1 = 4(x-3)^2 - 1$. Find $g(4)$.

a. 7

b. -1

c. -3

d. 3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Even and odd functions • Cross-Lesson: 6.01 Probability

Let $f(x) = x^2$ and $g(x) = -f(x-2) + 8 = -(x-2)^2 + 8$. Find $g(-8)$.

a. -96

b. -92

c. -88

d. 92

39 [Challenging] • Target: Domain restrictions • Cross-Lesson: 1.01 Complex Numbers

Find the domain of $g(x) = \frac{1}{x^2 - 144}$.

a. $14, -12$

b. $12, -12$

c. $-12, 12$

d. $12, -14$

40 [Challenging] • Target: Piecewise functions • Cross-Lesson: 2.01 Sequences & Series

Find the domain of $g(x) = \frac{1}{x^2 - 49}$.

a. $9, -7$

b. $7, -9$

c. $7, -7$

d. $-7, 7$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Rational Functions & Oblique Asymptotes • Cross-Lesson: 4.01 Matrices & Vectors

Find the oblique asymptote of $f(x) = \frac{3x^2 - 2x + 4}{x - 1}$.

a. $y = 3x + 1$

b. $y = 3x - 1$

c. $y = 3x + 4$

d. $y = 3x$

42 [Challenging] • Target: Rational Functions & Oblique Asymptotes • Cross-Lesson: 5.01 Introductory Calculus

What is the horizontal asymptote of $h(x) = \frac{5x - 2}{x^2 + 1}$?

a. $y = 0$

b. $y = 5$

c. $y = -2$

d. No horizontal asymptote

43 [Challenging] • Target: Piecewise & Absolute Value Graphs • Cross-Lesson: 6.01 Probability

What is the domain of $f(x) = \sqrt{9 - x^2}$?

a. $[-3, 3]$

b. $(-\infty, -3] \cup [3, \infty)$

c. $[0, 3]$

d. $[-9, 9]$

44 [Challenging] • Target: Inverse Trigonometric Graphs & Domains • Cross-Lesson: 1.01 Complex Numbers

What is the range of the principal $\arccos(x)$ function?

a. $[0, \pi]$

b. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. $[0, 2\pi]$

d. $(0, \pi)$

45 [Challenging] • Target: Transformations of Non-Linear Functions • Cross-Lesson: 2.01 Sequences & Series

How is the graph of $y = 2f(-x) + 3$ obtained from $y = f(x)$?

a. Reflect over y-axis, vertical stretch by 2, shift up 3

b. Reflect over x-axis, vertical stretch by 2, shift up 3

c. Reflect over y-axis, horizontal stretch by 2, shift up 3

d. Reflect over origin, shift up 3

© ClipSAT — clipsat.org | All Rights Reserved